# DSA Tutorial - 1

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**Topic: String Operations**

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## Input:

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#include <stdio.h>

int pallindrome(char input[]);
void copy(char input[], char output[]);
int stringlength(char input[]);
void substring(char input[]);
void reverse(char input[]);

int main() {
    int ch;
    char str[50];
    char output[50];
    printf(“Viraj T B25ET1053”);
    printf("Enter a string: ");
    scanf("%49s", str);
    printf("Choose an Operation:\n1. Check Palindrome\n2. Copy String\n3. Check String
Length\n4. Substring\n5. Reverse String\n");
    scanf("%d", &ch);

    switch(ch) {
        case 1: {
            int c = pallindrome(str);
            if (c == 1)
                printf("Palindrome\n");
            else
                printf("Not a Palindrome\n");
            break;
        }
        case 2:
            copy(str, output);
            break;
        case 3:
            printf("String length is: %d\n", stringlength(str));
            break;
        case 4:
            substring(str);
```

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            break;
        case 5:
            reverse(str);
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}

int stringlength(char input[]) {
    int i = 0;
    while(input[i] != '\0') {
        i++;
    }
    return i;
}

int pallindrome(char input[]) {
    int i = 0, j = stringlength(input) - 1;

    while(i < j) {
        if(input[i] != input[j]) {
            return 0;
        }
        i++;
        j--;
    }
    return 1;
}

void copy(char input[], char output[]) {
    int i;
    for (i = 0; input[i] != '\0'; i++) {
        output[i] = input[i];
    }
    output[i] = '\0';
    printf("Output string is: %s\n", output);
}

void substring(char input[]) {
    int i, position;
    char output[50];
    int n = stringlength(input);
```

```
    printf("Enter the position of substring: ");
    scanf("%d", &position);

    if(position >= 0 && position < n) {
        for(i = 0; i < n - position; i++) {
            output[i] = input[position + i];
        }
        output[i] = '\0';
        printf("Input string: %s\nOutput substring: %s\n", input, output);
    } else {
        printf("Position entered is out of range\n");
    }
}void reverse(char input[]) {
    int i;
    int n = stringlength(input);
    char output[50];

    for(i = 0; i < n; i++) {
        output[n - 1 - i] = input[i];
    }
    output[n] = '\0';
    printf("Input string: %s\nOutput (reversed) String: %s\n", input, output);
}
```

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## Output:

```
Viraj T B25ET1053
Enter a string: MMCOE123
Choose an Operation:
1. Check Palindrome
2. Copy String
3. Check String Length
4. Substring
5. Reverse String
5
Input string: MMCOE123
Output (reversed) String: 321EOCMM


=== Code Execution Successful ===
```